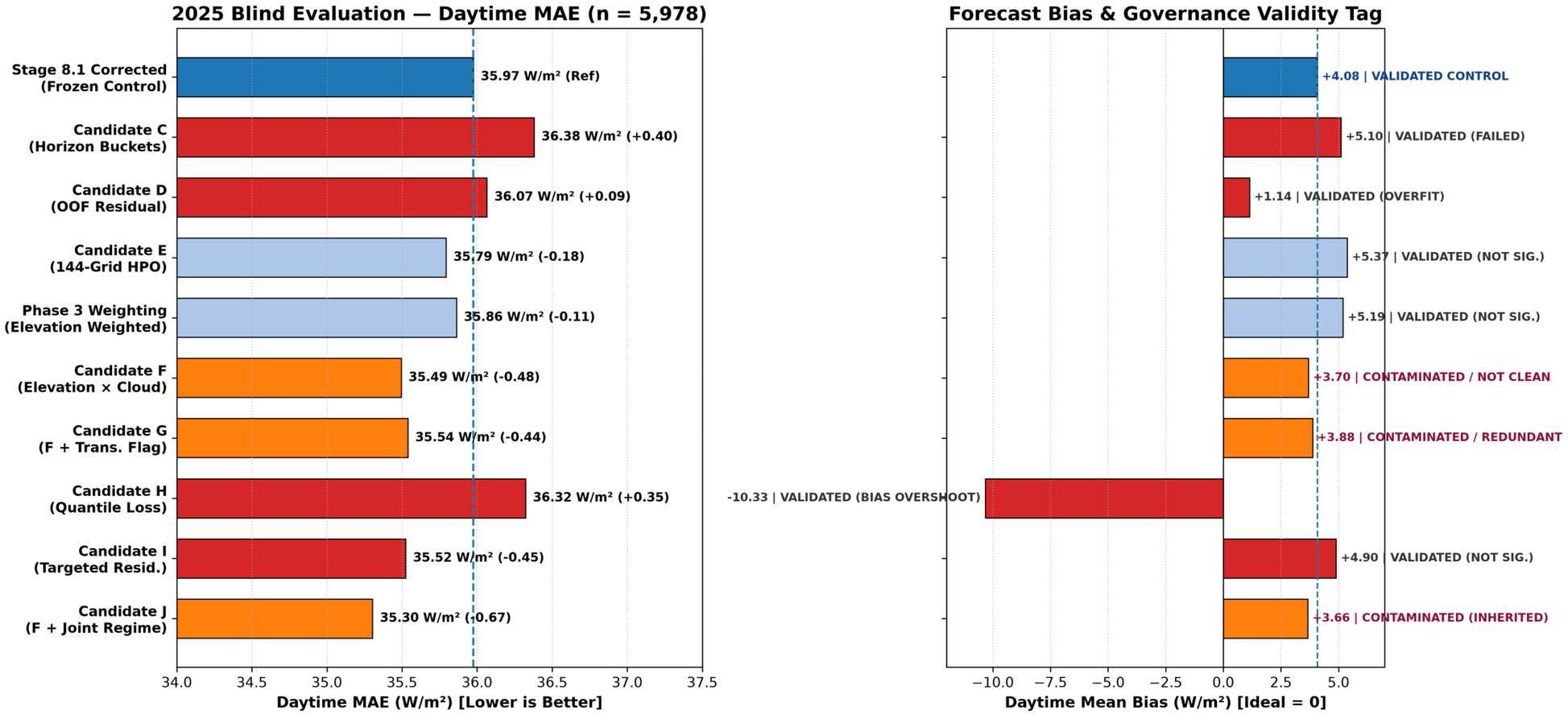
GRAPH A: POLARIS Solar Forecasting Model Leaderboard (2025 Blind Holdout)

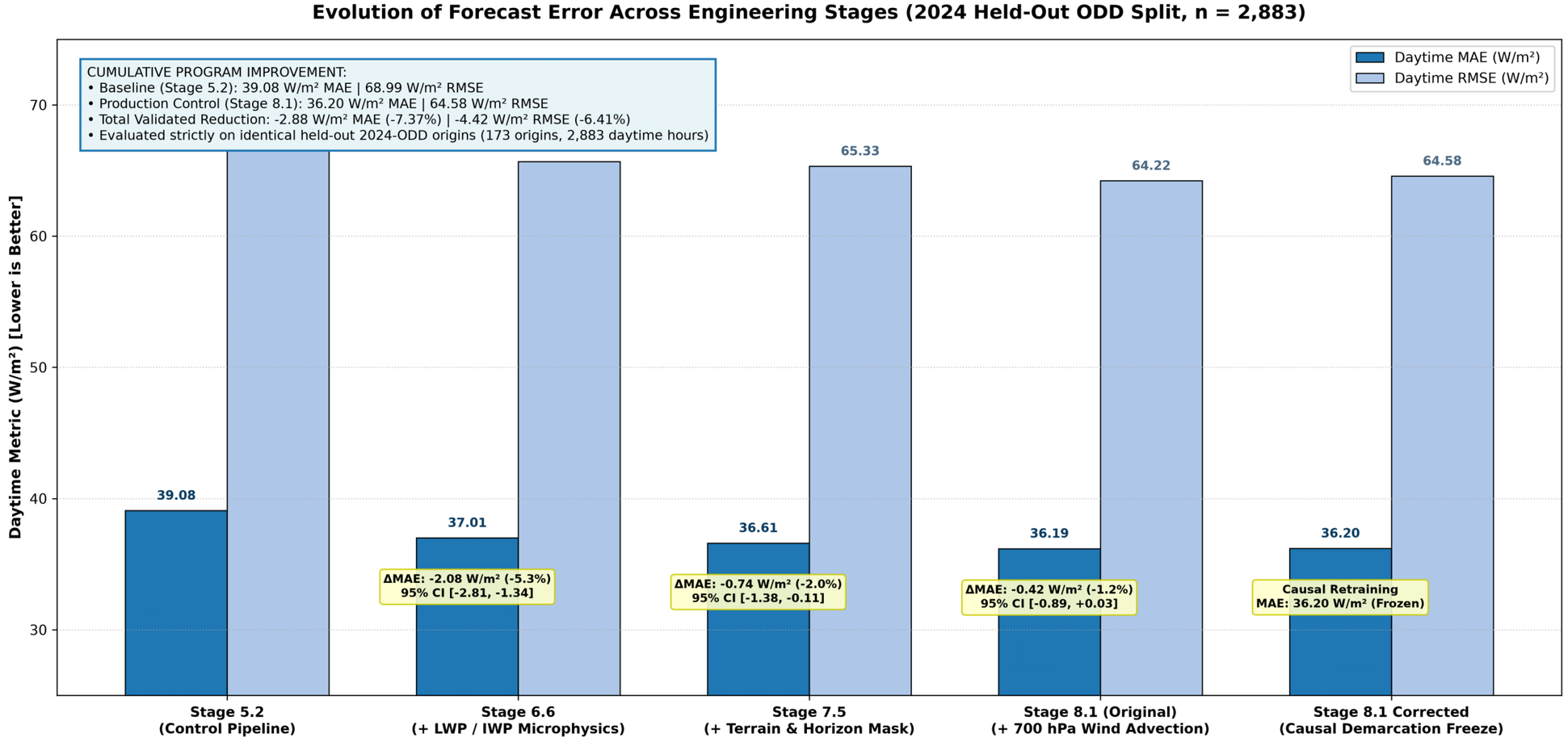


GOVERNANCE DECISION: winner: null | No model promoted to production.
Candidate F & J improvements (-0.48 to -0.67 W/m²) are contaminated by 2025-EVEN ensemble parameters fitted on realized 2025 targets.
Source: stage7/stage8/stage8_1_station_tuning_research/ (CANDIDATE_RESULTS.json, TARGETED_CANDIDATE_F_CAUSALITY_TRAINING_INTEGRITY_AUDIT.md)

PRESENTATION TALKING POINT: On the strictly held-out 2025 blind evaluation (n = 5,978), the frozen Stage 8.1 Corrected control achieves 35.97 W/m² MAE. Candidates C through E fail to produce statistically significant blind gains.

CAVEAT & GOVERNANCE BOUNDARY: Candidate F (-0.48 W/m²) and Candidate J (-0.67 W/m²) show lower MAE point estimates, but their reported blind evaluations are CONTAMINATED by 2025-EVEN ensemble parameters fitted on realized 2025 solar irradiance targets. Governed status is strictly winner: null.

GRAPH B: Measured Accuracy Improvements Across POLARIS Project Stages

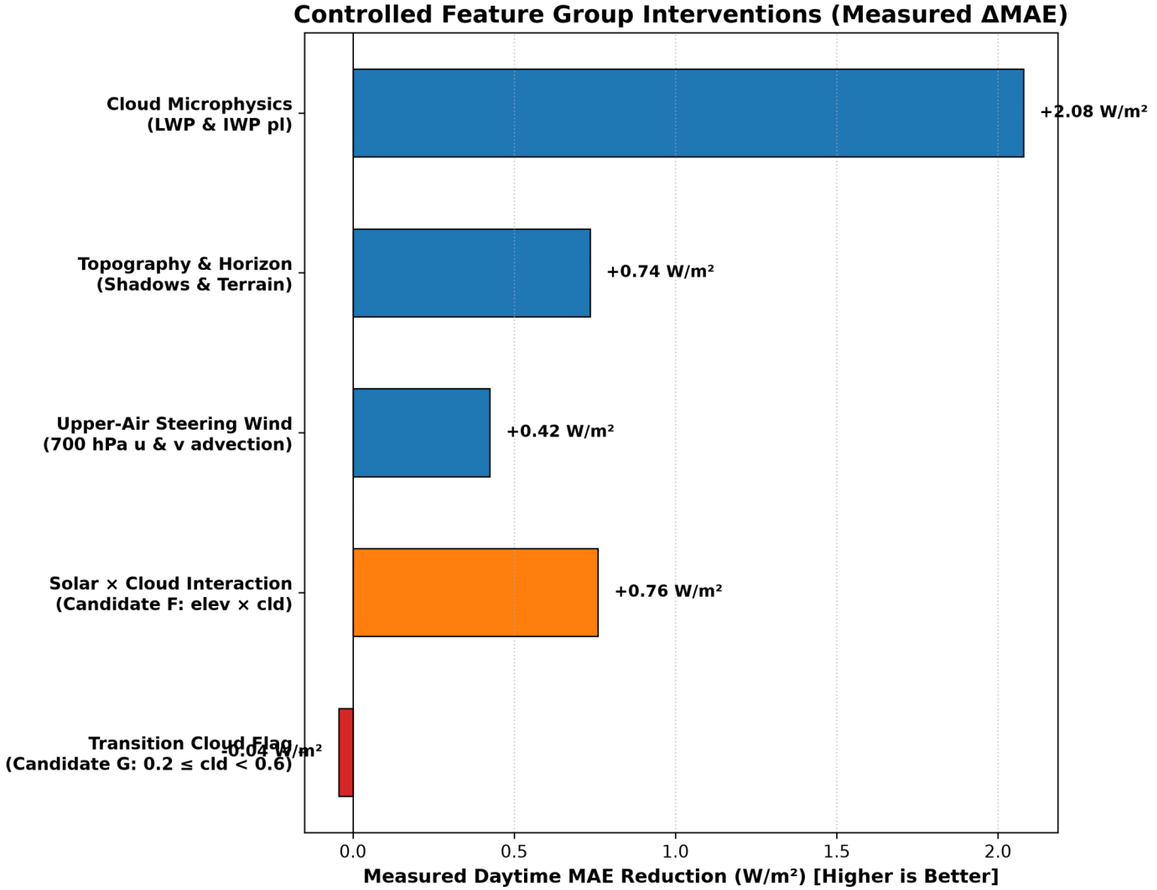


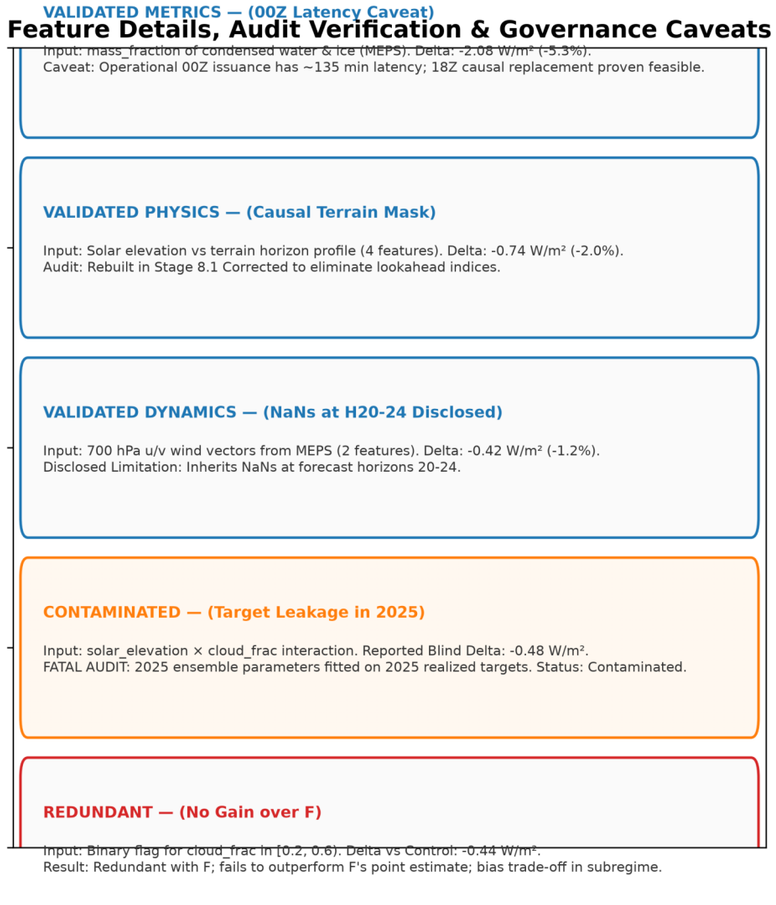
Sources: stage6/stage6_6_lwp_iwp/odd_evaluation_results.json, stage7/stage7_5_terrain_horizon/stage7_5_official_odd_results.json, stage7/stage8/stage8_1_700hpa_wind/stage8_1_official_odd_results.json, stage7/stage8/stage8_1_corrected_2025/stage8_1_CORRECTED_official_odd_results_2024.json

PRESENTATION TALKING POINT: Daytime MAE was reduced from 39.08 W/m² (Stage 5.2) to 36.20 W/m² (Stage 8.1 Corrected), representing a verified -2.88 W/m² (-7.37%) error reduction and a -4.42 W/m² (-6.41%) RMSE reduction on identical held-out origins.

CAVEAT & GOVERNANCE BOUNDARY: Stages 6.6 and 7.5 historical numbers reflect preliminary feature extractors later audited and hardened in Stage 8.1 Corrected to eliminate lookahead indices and restore strict causal boundaries.

GRAPH C: POLARIS Feature Engineering Impact and Scientific Validity



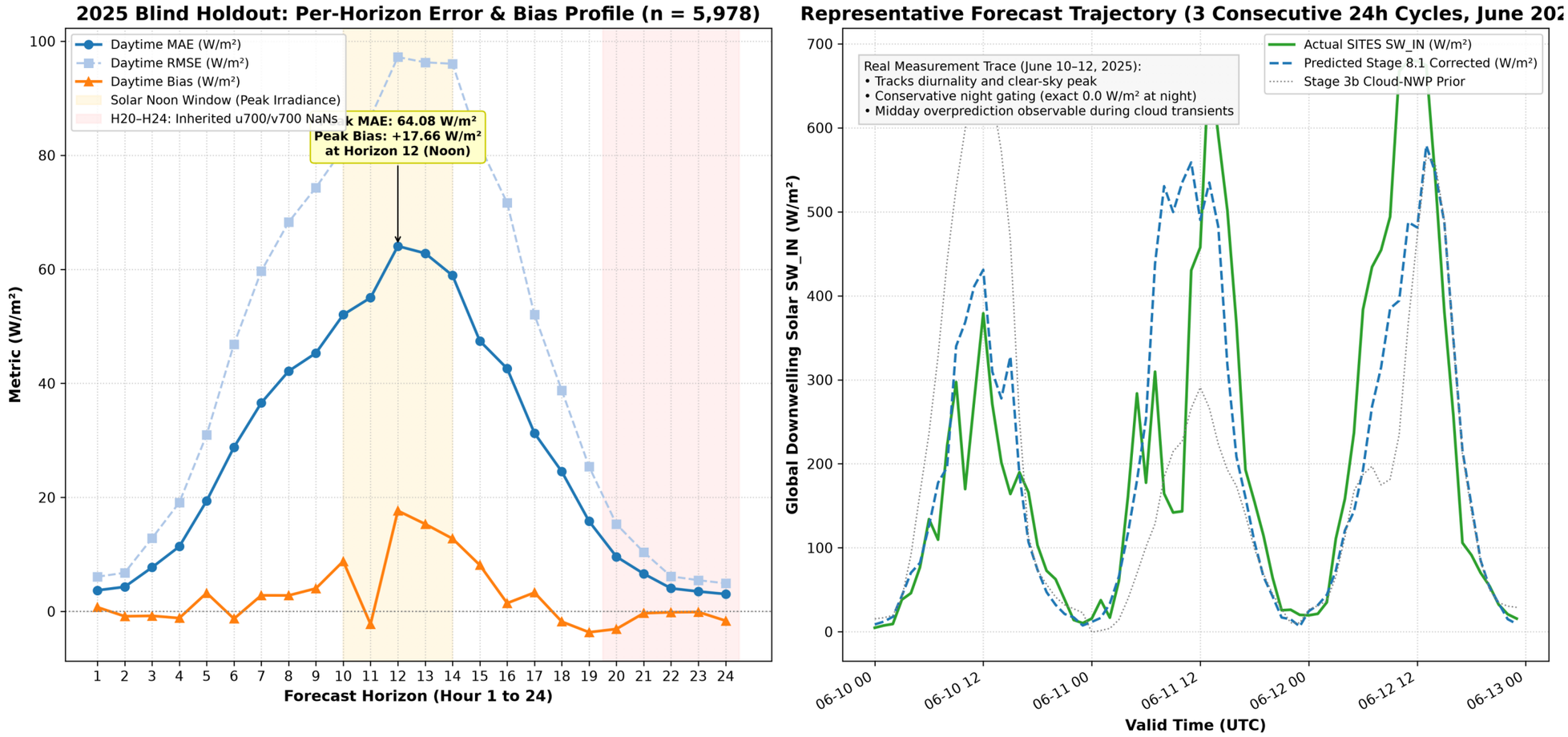


Key Safeguard: Do not attribute causal importance to multi-change experiments. Tagged strictly per repository audit findings.
Source: stage6_6 odd_evaluation_results.json, stage7_5 official_odd_results.json, stage8_1 official_odd_results.json, TARGETED_CANDIDATE_F_CAUSALITY_TRAINING_INTEGRITY_AUDIT.md

PRESENTATION TALKING POINT: Isolated physical interventions demonstrate clear value: Microphysics (LWP/IWP) contributed -2.08 W/m² (-5.3%), Topography & Horizon shadows contributed -0.74 W/m² (-2.0%), and 700 hPa Wind Advection contributed -0.42 W/m² (-1.2%).

CAVEAT & GOVERNANCE BOUNDARY: MEPS 00Z LWP/IWP source has operational publication latency (~135 min after cutoff); 18Z causal replacement is feasible. Candidate F elevation × cloud interaction (-0.76 W/m² ODD) cannot be certified clean on 2025 blind data due to target leakage.

GRAPH D: POLARIS Solar Forecasting Performance Across Horizons and Time

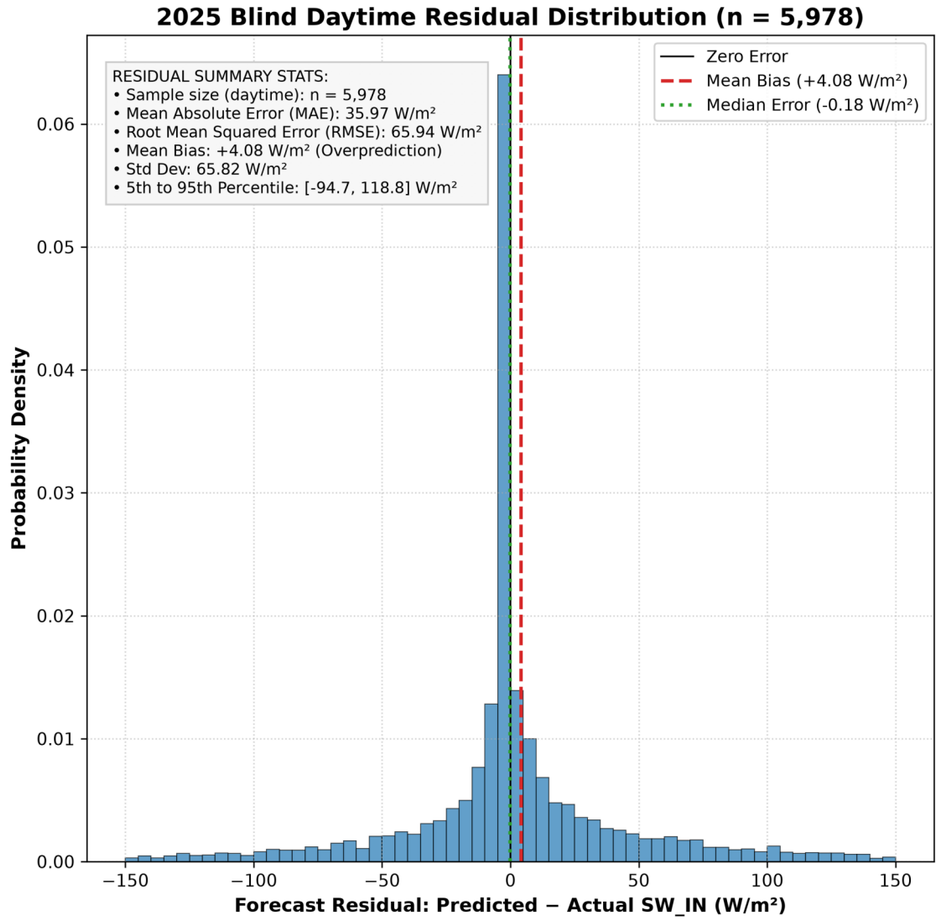


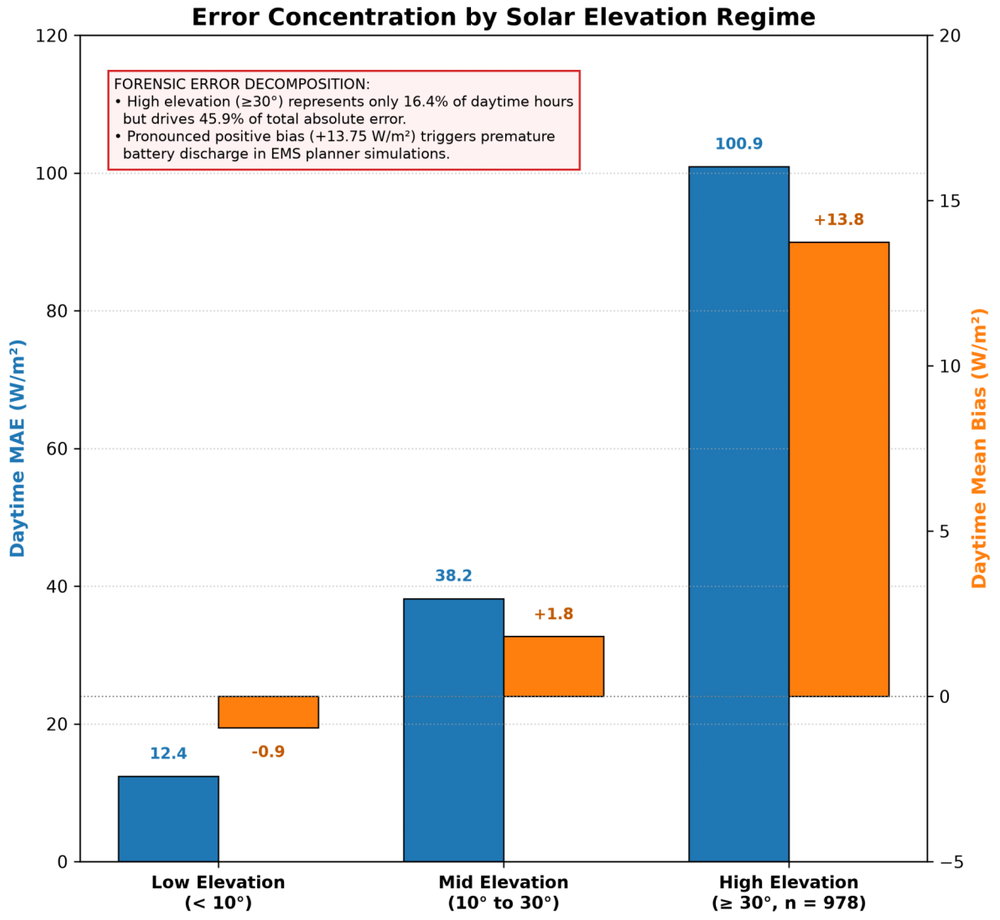
Disclosed Limitation: MEPS u700/v700 features carry inherited NaNs at horizons 20-24 due to upstream lead-time truncation.
Source: stage7/stage8/stage8_1_corrected_2025/stage8_1_CORRECTED_2025_blind_horizon_metrics.csv & stage8_1_CORRECTED_2025_blind_predictions.csv

PRESENTATION TALKING POINT: Errors track solar intensity diurnally: MAE rises from ~3.7 W/m² at H1 to 64.08 W/m² at solar noon (H12), with a peak midday overprediction bias (+17.66 W/m²). Real measurements demonstrate accurate dawn/dusk transitions and strict zero-night gating.

CAVEAT & GOVERNANCE BOUNDARY: DISCLOSED LIMITATION: MEPS u700/v700 features contain inherited NaNs at forecast horizons 20–24 due to upstream archive truncation, routed by native GBDT missing-value branching.

GRAPH E: Forecast Residual Distributions & Solar Elevation Regime Analysis



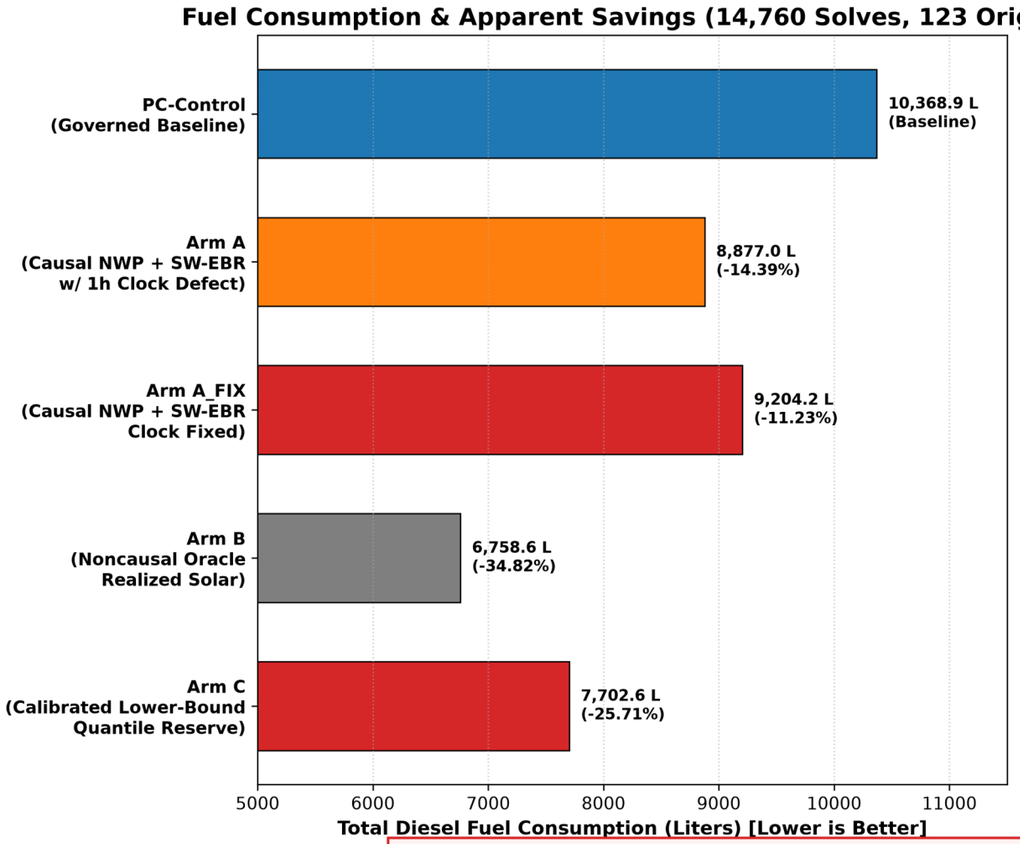


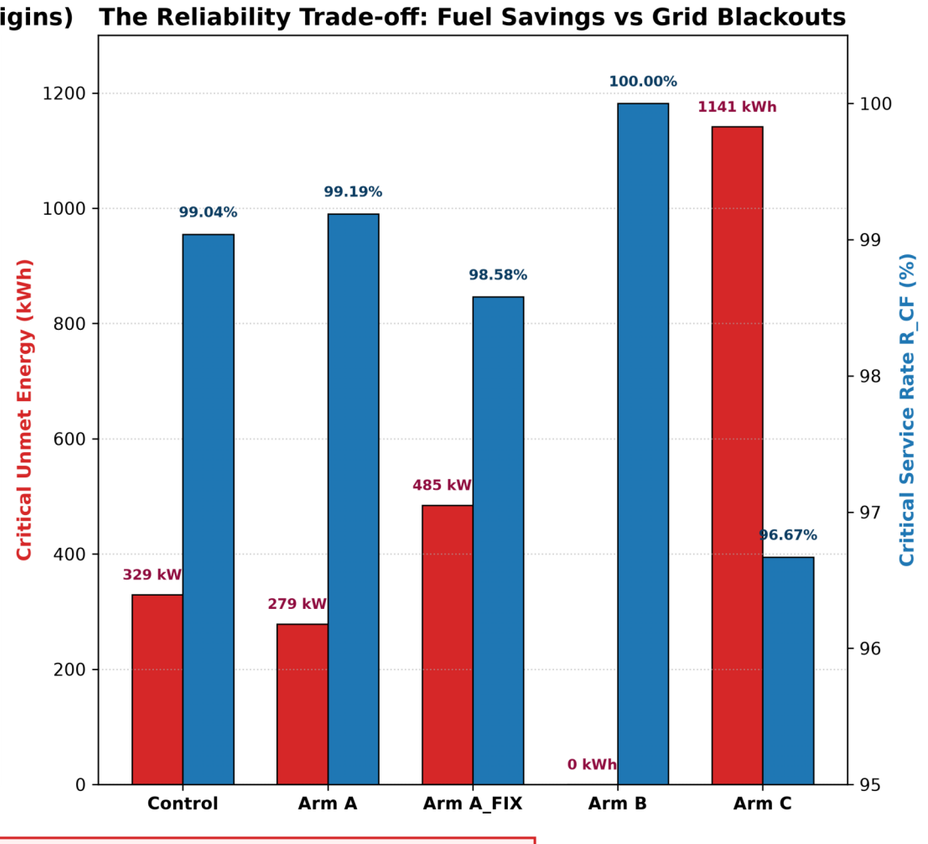
Source: stage7/stage8/stage8_1_corrected_2025/stage8_1_CORRECTED_2025_blind_predictions.csv & CANDIDATE_F_RESULTS.json
Forensics: stage7/stage8/stage8_1_station_tuning_research/ERROR_FORENSICS_REPORT.md

PRESENTATION TALKING POINT: Daytime residuals on 2025 blind data exhibit a mean overprediction bias of +4.08 W/m² (MAE 35.97 W/m²). Elevation regime decomposition reveals that 45.9% of total absolute error and massive positive bias (+13.75 W/m²) are concentrated in high-elevation hours (≥30°).

CAVEAT & GOVERNANCE BOUNDARY: Interventions attempting to force lower bias at high elevation (Candidates H & I) overshot into severe underprediction (-10.33 W/m² overall, -21.67 W/m² high elevation) or overfit validation data. Candidate F reduced high-elevation MAE to 98.68 W/m² but carries target contamination.

GRAPH F: POLARIS Optimization and Diesel Reduction (EXP-THREE-ARM-2024)





CRITICAL SCIENTIFIC & GOVERNANCE SAFEGUARDS:
1. Current Governed Status: winner: null (RESULTS.json winner_note: No arm is promotion-eligible).
2. Arm A (14.39% saving) carries a 1-hour solar position clock defect inflating savings; 22.8% of origins used MORE fuel than control.
3. Arm A_FIX isolates the defect: fuel savings drop to 11.23% and reliability gates S1, S2, S4 all FAIL.
4. Arm B (34.82% saving) is NONCAUSAL (consumes future realized solar; unrealizable at real decision time) and fails strict paired gate S4.
5. Arm C (25.71% saving) suffers catastrophic reliability collapse: critical unmet energy increases by 3.5× (1,141 kWh vs 329 kWh).

PRESENTATION TALKING POINT: 14,760 optimizer solves across 123 origins demonstrate that netting solar forecasts against battery reserve (EBR) reduces diesel fuel. However, fuel savings alone do NOT prove operational readiness: reliability must be co-evaluated.

CAVEAT & GOVERNANCE BOUNDARY: CRITICAL GOVERNANCE FACTS: (1) Current status: winner: null. (2) Arm A's 14.39% saving is inflated by a 1h clock defect. (3) Fixing the clock (Arm A_FIX) reduces savings to 11.23% and fails reliability gates S1, S2, S4. (4) Arm B's 34.82% saving is noncausal oracle foresight. (5) Arm C causes a 3.5× surge in critical unmet demand.

POLARIS Pipeline Evolution, Causal Boundaries, and Current Governed Status

1. BASELINE NWP (Stages 3–5)
- metno_seamless 6 vars
- Clear-sky & sun geometry
- HistGradientBoosting
- Daytime MAE: 39.08 W/m²
- Status: Initial Control

→

2. MICROPHYSICS (Stage 6.6)
- Liquid/Ice Water Path
- MEPS pressure level
- Daytime MAE: 37.01 W/m²
- ΔMAE: -2.08 W/m² (-5.3%)
- Disclosed: 00Z Latency

→

3. TOPOGRAPHY & WIND (Stages 7.5 & 8.1)
- Horizon terrain mask (4)
- 700 hPa wind advection (2)
- 37 feature contract
- Daytime MAE: 36.19 W/m²
- Disclosed: H20-24 NaNs

→

4. CAUSAL REPAIR FREEZE (Stage 8.1 Corrected)
- Causal index correction
- Retrained EVEN-2024
- Daytime MAE: 36.20 W/m²
- 2025 Blind: 35.97 W/m²
- Status: FROZEN CONTROL

→

5. CANDIDATE F & SUCCESSORS (Station Tuning)
- solar_elev × cloud_frac
- Reported Blind: 35.49 W/m²
- ΔMAE: -0.48 W/m² [CI excl 0]
- BUT: 2025 target leakage
- Status: CONTAMINATED

A. MEPS LWP/IWP ISSUANCE AUDIT
- Finding: MEPS 00Z run initializes at same instant as forecast cutoff.
- Reality: Atmospheric processing latency is ~135 min (MetCoOp documentation).
- Remediation: Evaluated prior-day 18Z product; contains identical LWP/IWP fields.
- Status: 18Z causal replacement proven technically & temporally feasible.

B. 2025 TARGET CONTAMINATION FIREWALL
- Audit Finding: stage4_results_2025/ensemble_parameters.json (k_h/w_h) were fitted on realized 2025 actual_SW_IN solar irradiance targets!
- Impact: 4 derived features inherited this target leakage for all 2025 rows.
- Consequence: Candidate F blind -0.48 W/m² improvement is NOT clean blind proof.
- Governed Verdict: TRAINING_CONTAMINATION_VERIFIED.

C. OPTIMIZER RELIABILITY FIREWALL
- EXP-THREE-ARM-2024 tested solar EBR reserve netting on 14,760 solves.
- Arm A reported 14.39% fuel savings but had 1h clock defect.
- Clock-corrected Arm A_FIX dropped to 11.23% savings and FAILED reliability.
- Arm B saved 34.82% but used noncausal oracle solar (unrealizable).
- Governed Verdict: winner: null. No arm promotion-eligible.

CURRENT AUTHORITATIVE STATUS & PRESCRIBED ROADMAP
- PRODUCTION BASELINE: Frozen Stage 8.1 Corrected model (37 features, MAE = 36.20 W/m² on 2024-ODD, 35.97 W/m² on 2025-Blind) remains the sole control.
- CURRENT GOVERNED STATUS: winner: null across all candidate models and optimizer arms. No promotion is authorized.
- COMPLETED ENGINEERING: Candidate F successor dataset v2 has passed an independent dataset audit (clean 18Z features, zero 2025 target leakage).
- CLEAR PROVENANCE BOUNDARY: Dataset audit passing does NOT mean the successor model has been trained or evaluated.
- NEXT SAFE ACTION: Model training requires its own explicit, separate governance authorization; 2025 blind set must remain strictly held out.

PRESENTATION TALKING POINT: POLARIS enforces institutional integrity: independent dataset audits, origin-level bootstrap CI testing, and causality firewalls prevent premature deployment of flawed models or unsafe optimizer policies.

CAVEAT & GOVERNANCE BOUNDARY: The Candidate F successor v2 dataset has passed independent audit (clean 18Z features, zero 2025 leakage), but the successor model HAS NOT BEEN TRAINED OR EVALUATED. Governed status winner: null is maintained until formal promotion review.